STREAMING DATA MANAGEMENT AND TIME SERIES ANALYSIS PROJECT

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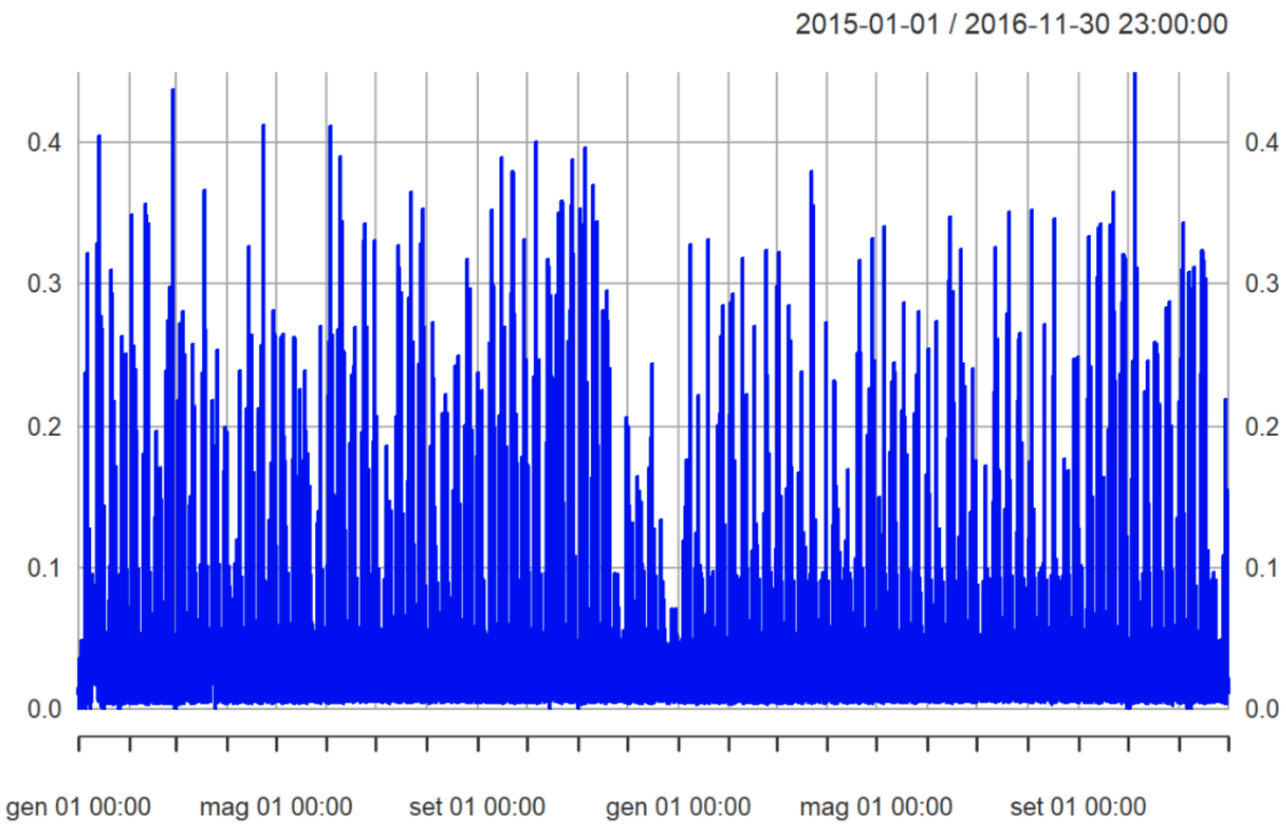
AIM OF THE PROJECT

The goal of this project is to analyze and forecast road traffic flows using advanced time series modeling techniques.



DATA OVERVIEW

Dataset: Hourly traffic data from Jan 1, 2015 to Nov 30, 2016 (17,544 observations).

Prediction Target: Full December 2016 (744 hourly forecasts).

Distribution: range 0–0.45, mean 0.046, median 0.037, SD 0.049.

DATA PRE-PROCESSING

01

ZEROS AND MISSING VALUES

The time series doesn't contain any missing vaulues but it contains 107 zero values, which were imputed using hourly medians.

02

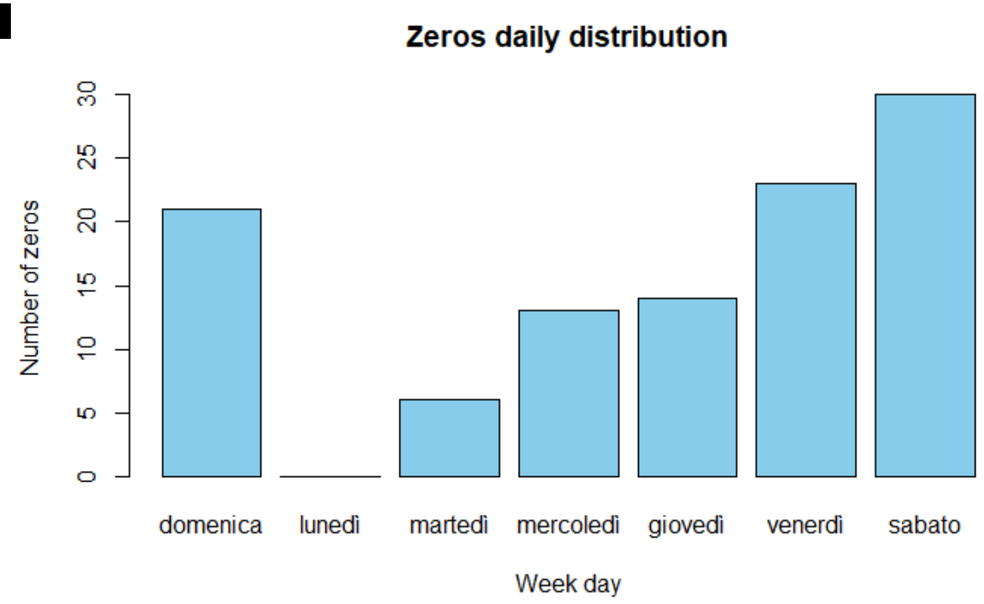
OUTLIERS

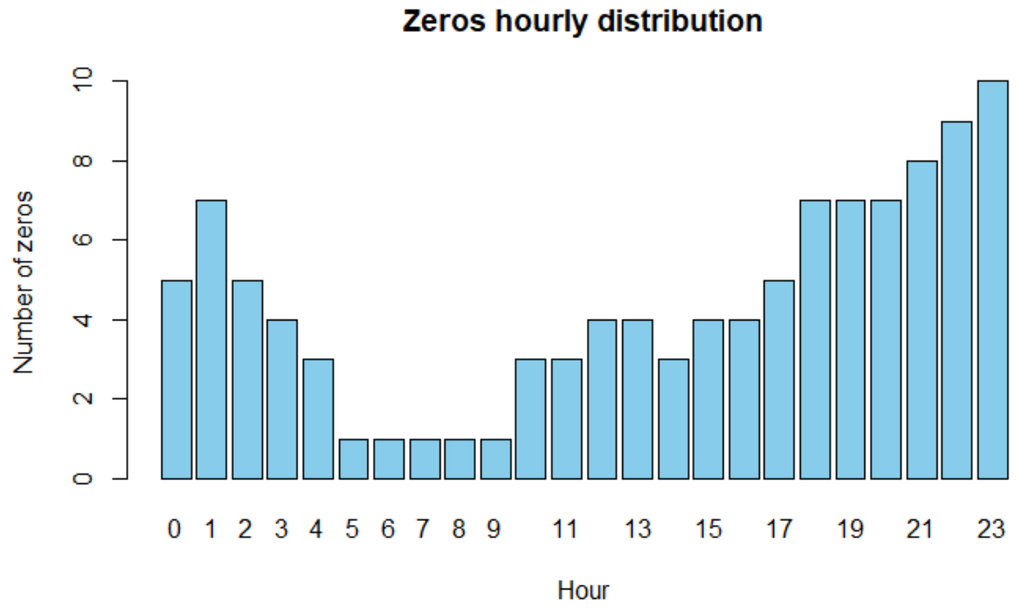
A total of 168 outliers were identified (above the 99th percentile), and replaced with the global median value. This helps reduce their impact on model performance.

03

TIME SERIES SPLITTING AND STATIONARITY

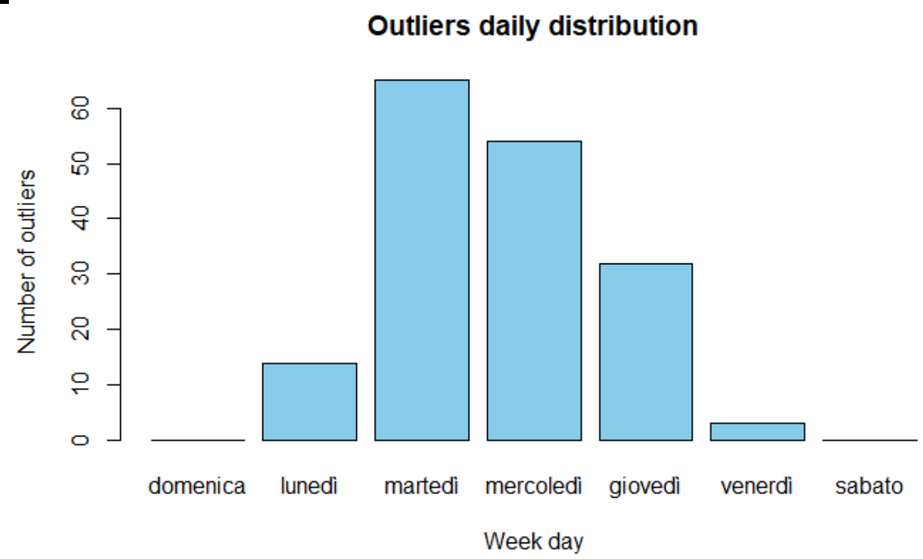
The original time series was split into 24 separate time series, one for each hour of the day. A Box-Cox transformation was applied to each of them using the corresponding optimal lambda value.

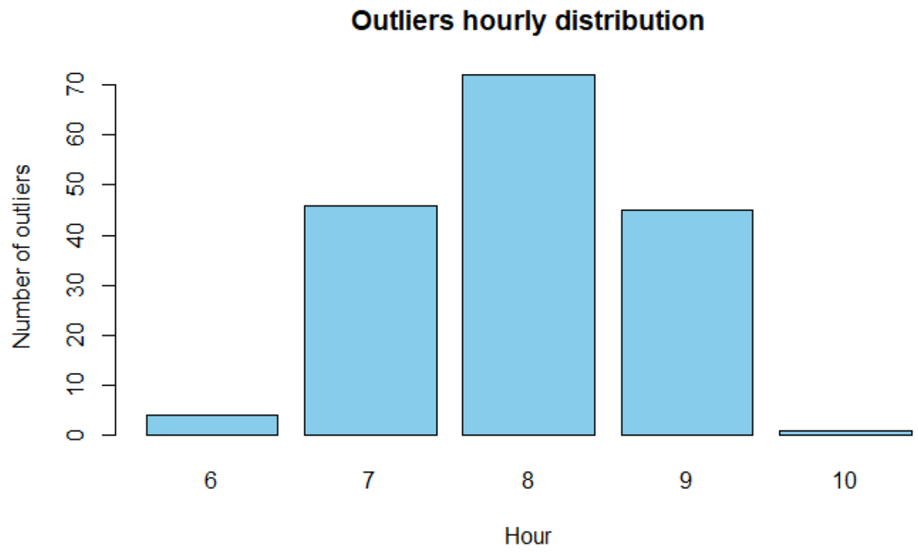




ZEROS

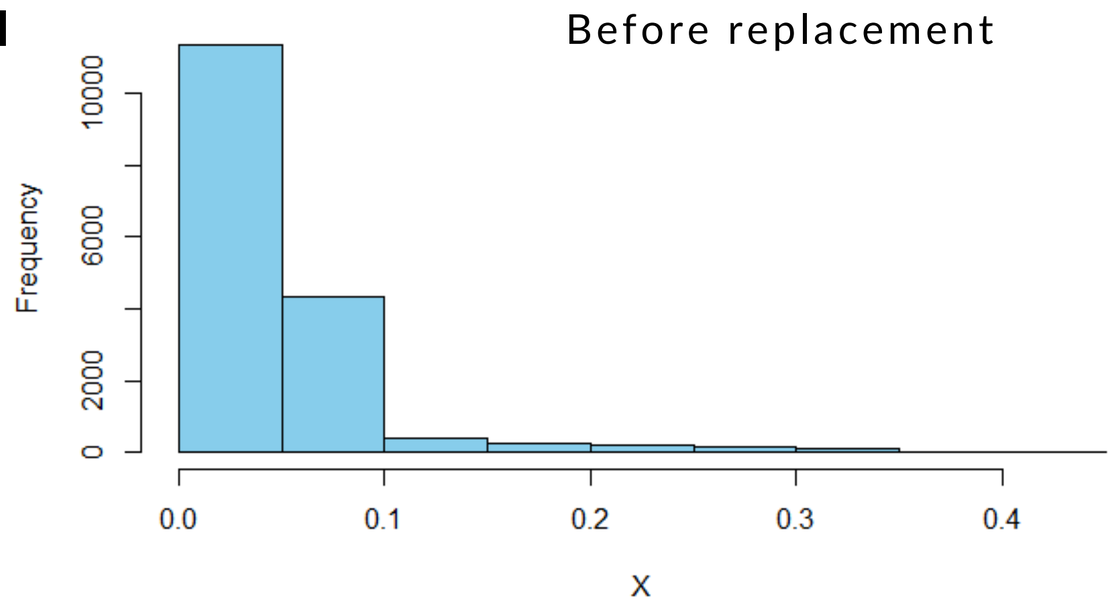
This imputation was based on the assumption that a complete traffic shutdown is highly unlikely. Moreover, the majority of zero values are concentrated after 5:00 PM and throughout the night, predominantly on weekends and occasionally on Fridays.

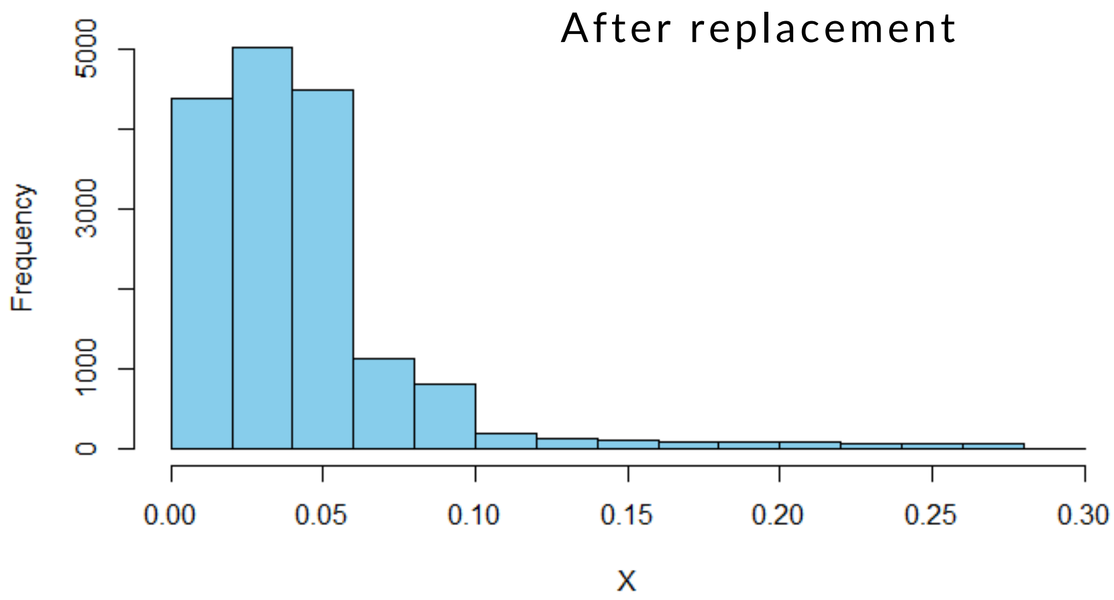




OUTLIERS

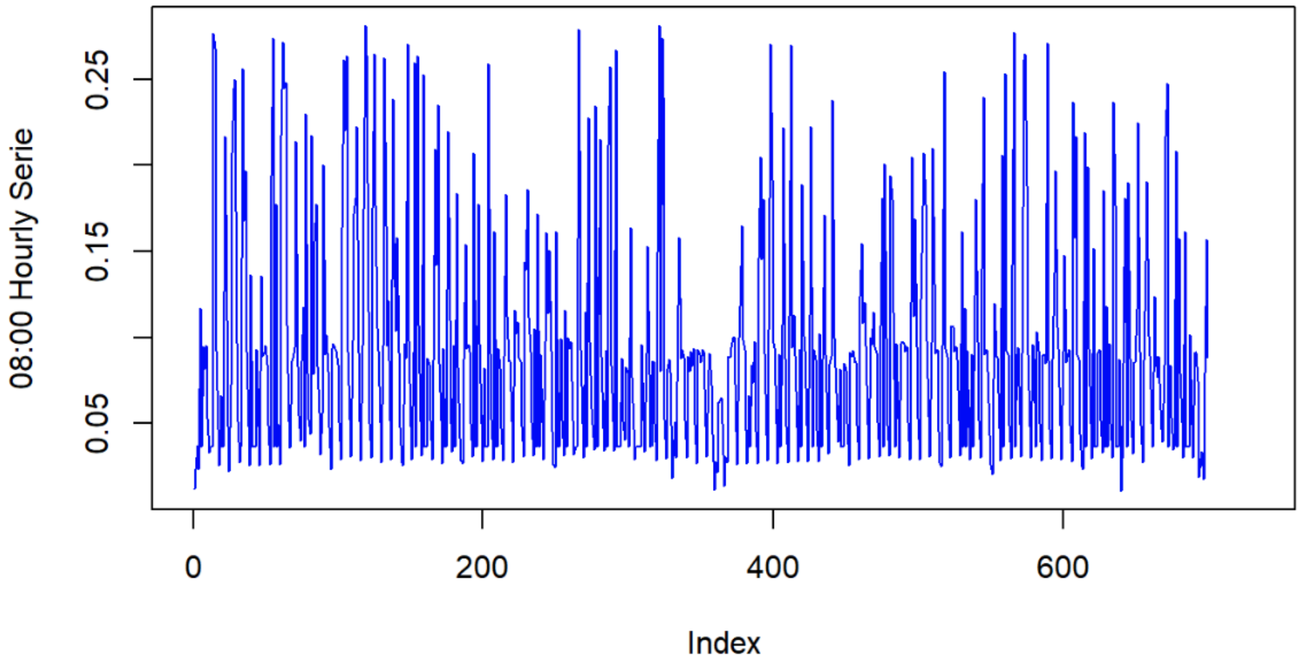
Although the hourly and daily distribution of these outliers suggests they occur during plausible periods, such as peak morning hours, they were still replaced to help the models focus on learning general patterns without being influenced by extreme, infrequent spikes.





OUTLIERS

The traffic measurements are mostly concentrated below 0.1, with very few values exceeding 0.3. These high values, likely belonging to a low-frequency class, are considered outliers and replaced with the global median (0.0368).



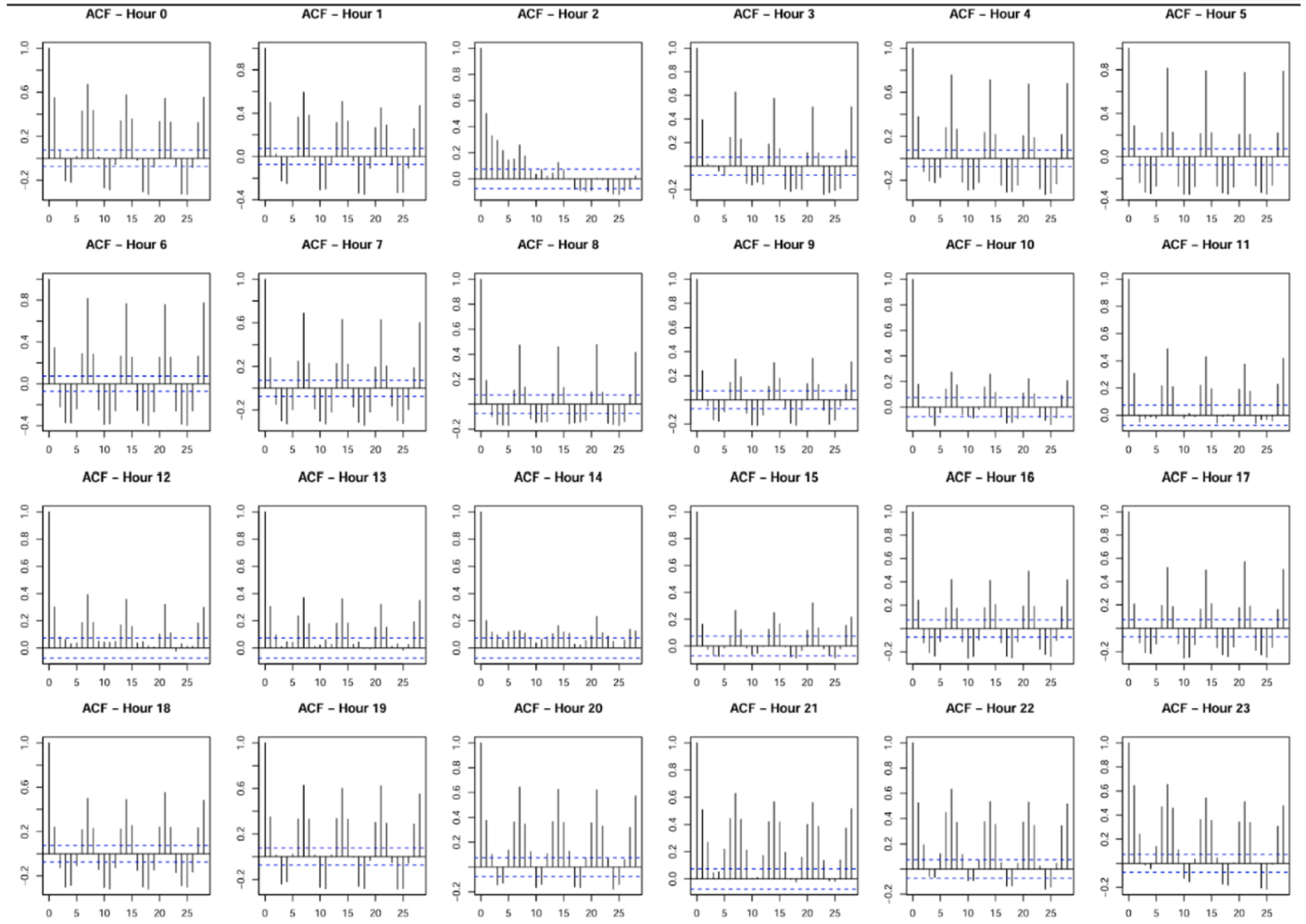
TIME SERIES SPLITTING

The original time series was then split into 24 separate time series, one for each hour of the day. This transformation allows the models to focus on learning the specific traffic patterns associated with a particular hour.

λ	Transformation
-2	$\frac{1}{x^2}$
-1	$\frac{1}{x}$
-0.5	$\frac{1}{\sqrt{x}}$
0	$\log(x)$
0.5	$\sqrt{x}$
1	x
2	x^2

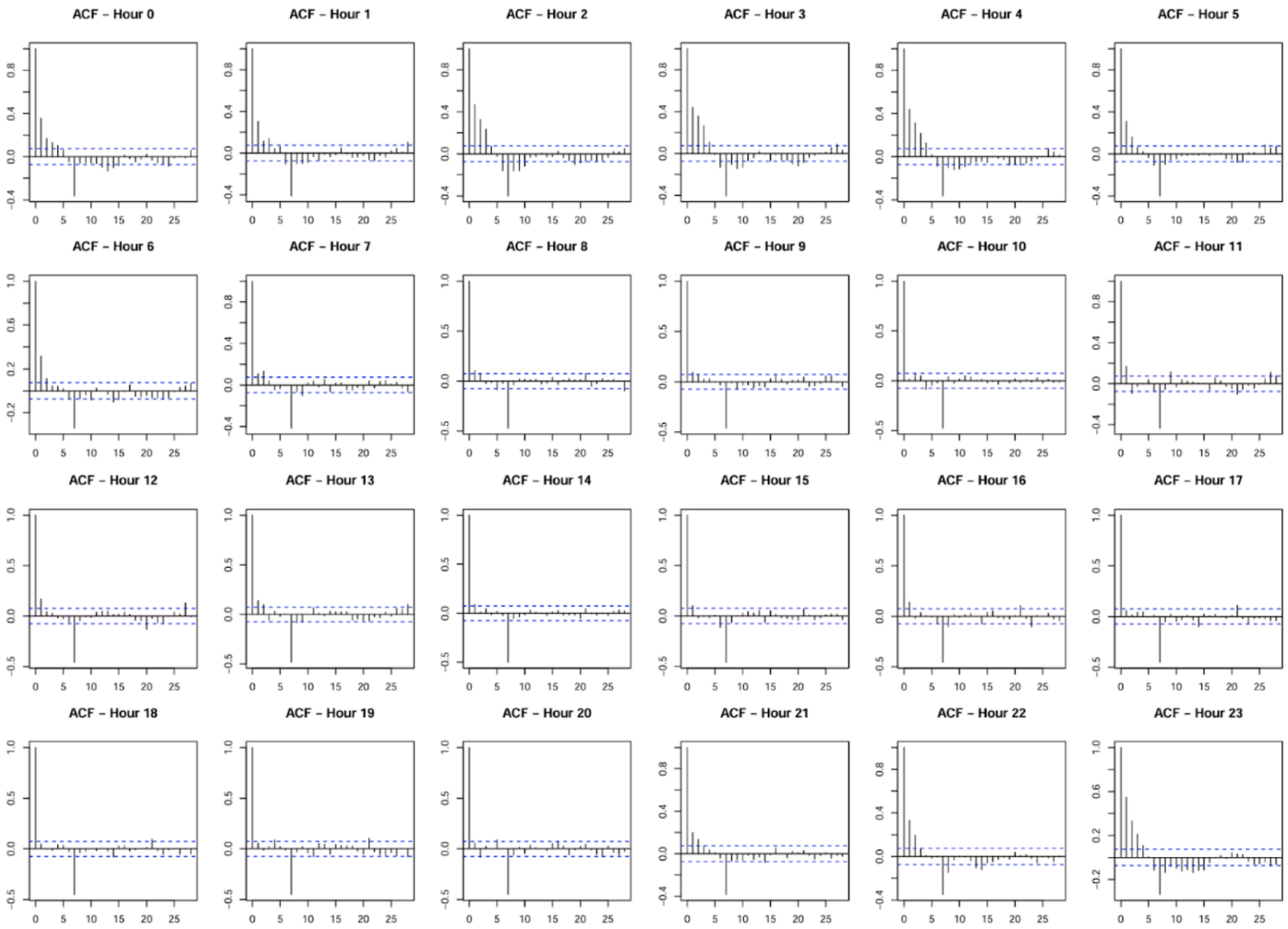
STATIONARITY

For each of the 24 hourly series, the optimal value of the Box-Cox lambda parameter was computed to identify the most appropriate transformation.



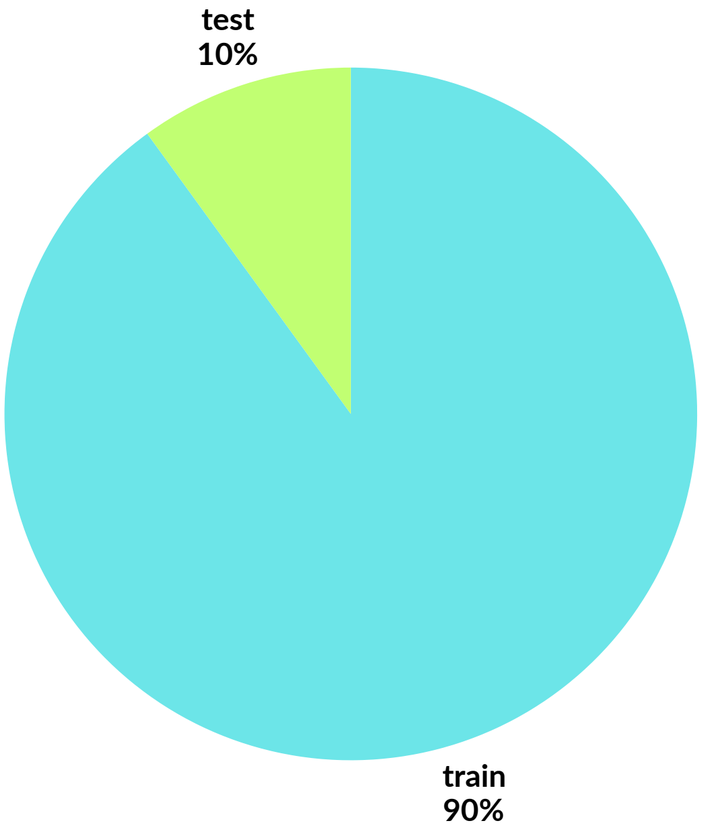
SEASONALITY

After applying the Box-Cox transformations, seasonality was further analyzed by plotting the Autocorrelation Function (ACF) which suggest a weekly seasonality (7-day) in nearly all the series, with the exception of hours 2 and 14, where this pattern is not clearly observed.



SEASONALITY

A seasonal differencing of period 7 was applied to all series to remove this recurring weekly component.
Later the Augmented Dickey-Fuller (ADF) test was performed to assess stationarity in mean.



TRAIN & TEST DATASETS

Each of the 24 hourly time series was then split into a training set and a test set. The training set spans from January 1, 2015, to September 21, 2016, and the test set covers the period from September 22 to November 30, 2016.

HOLIDAYS DUMMY VARIABLES

A dataset containing dummy variables for major holidays within the considered period.

- Christmas Day
- Christmas Eve
- St. Stephen’s Day
- New Year’s Day
- New Year’s Eve
- Epiphany (January 6th)
- Valentine’s Day
- Easter
- Easter Saturday
- Easter Monday
- Easter Tuesday
- Ferragosto (August 15th)
- Day of the week
- Day of the year

ARIMA MODELS

The three most promising ARIMA models were selected and further refined by incorporating holiday dummy variables. Their resulting performances (the average MAE across all 24 series) were then compared.

Without dummies

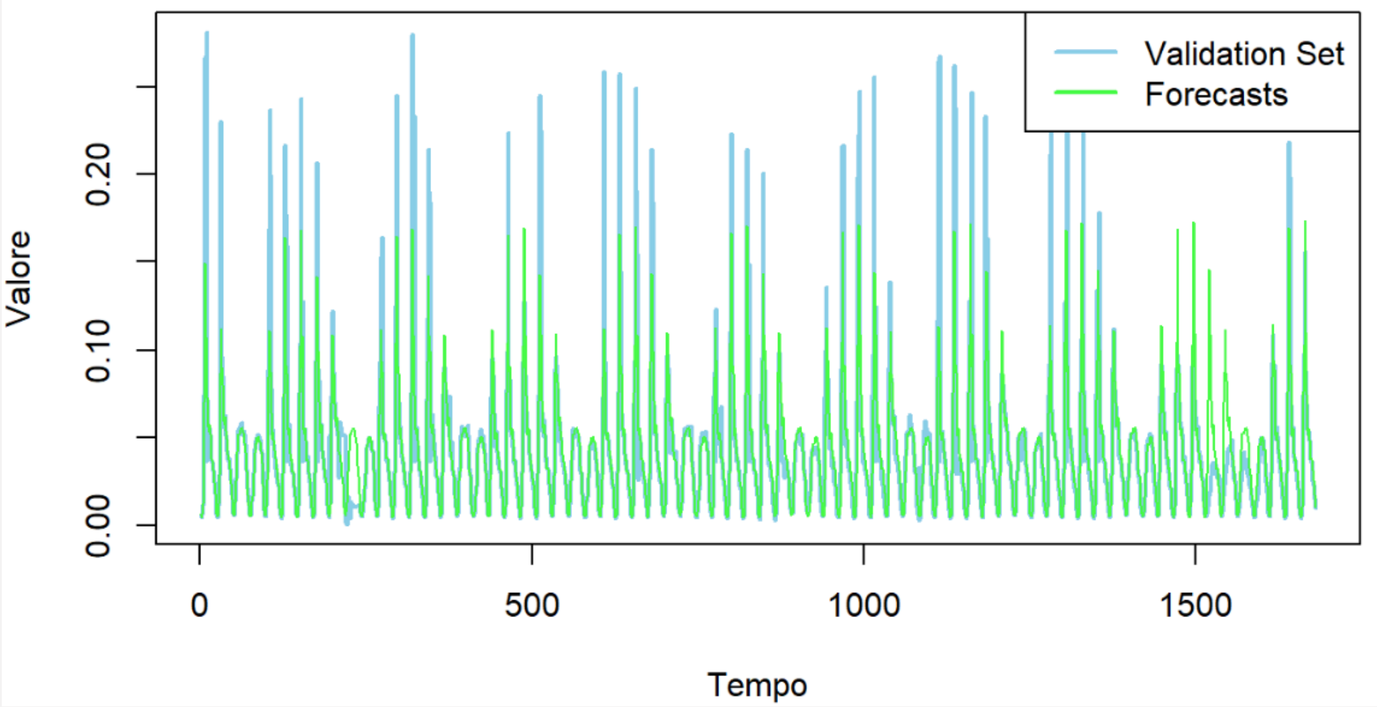
Model	MAE
ARIMA (3,1,1) (0,1,1) (7)	0.0091626
ARIMA (3,1,1) (0,1,1) (7)	0.0091629
ARIMA (3,1,1) (0,1,1) (7)	0.009155

With dummies

Model	MAE
ARIMA (3,1,1) (0,1,1) (7)	0.009141
ARIMA (3,1,1) (0,1,1) (7)	0.009170
ARIMA (3,1,1) (0,1,1) (7)	0.009166

BEST ARIMA MODEL

ARIMA (3,1,1) (0,1,1) (7) with dummies



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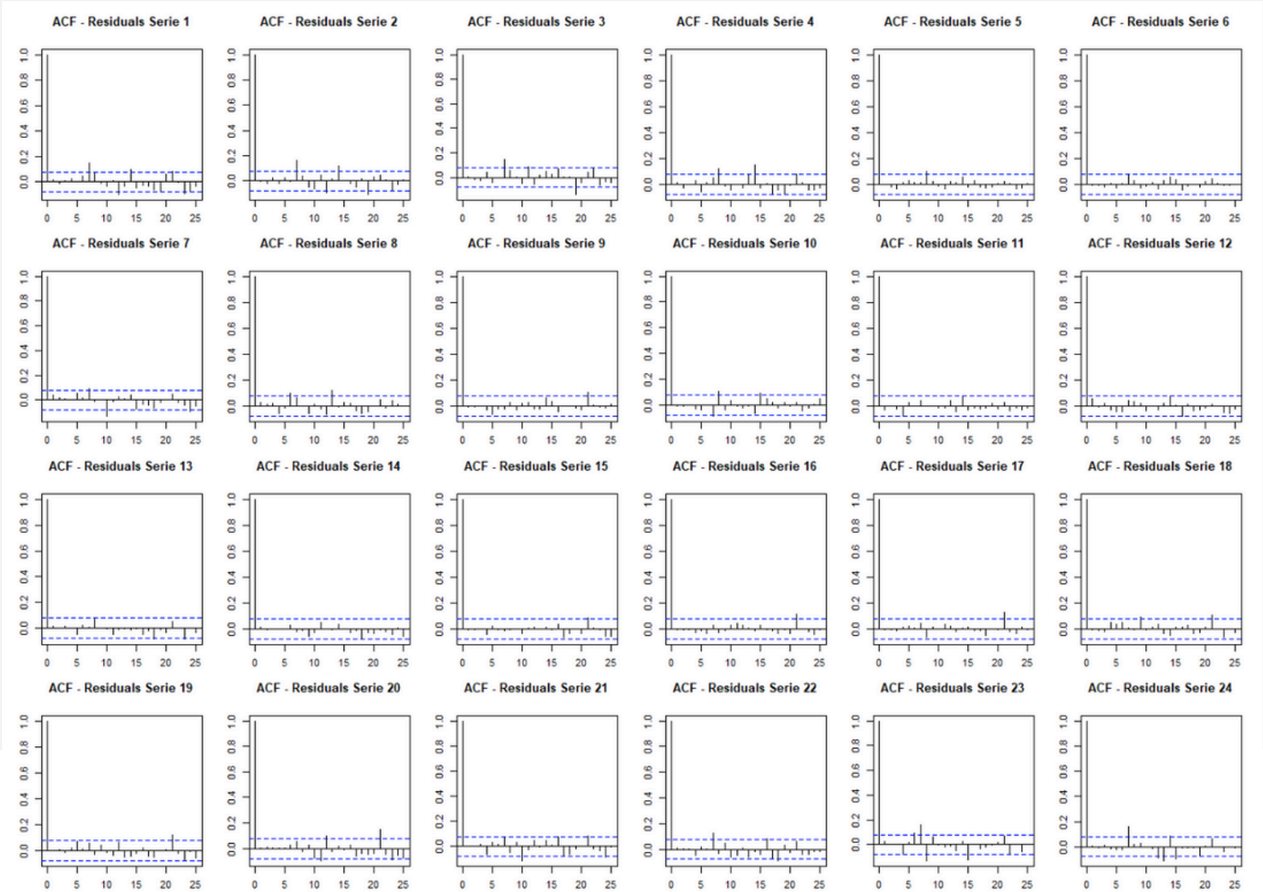
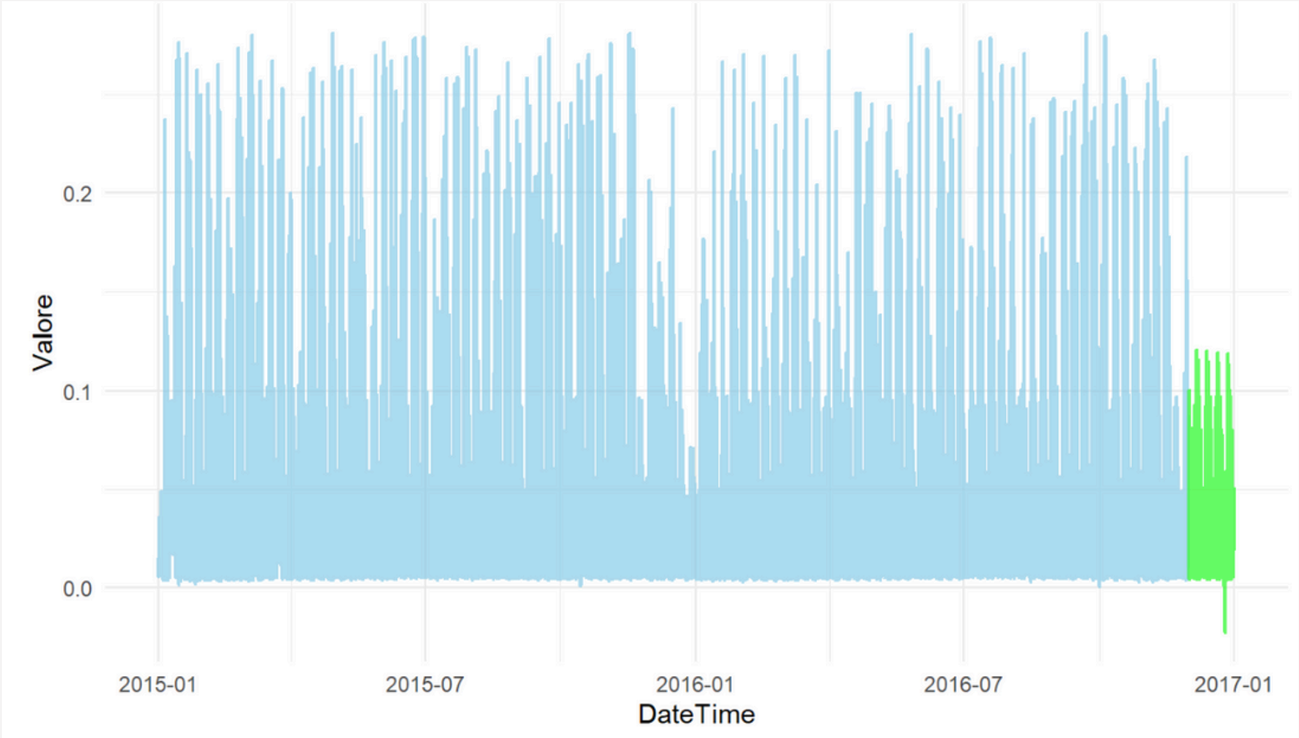


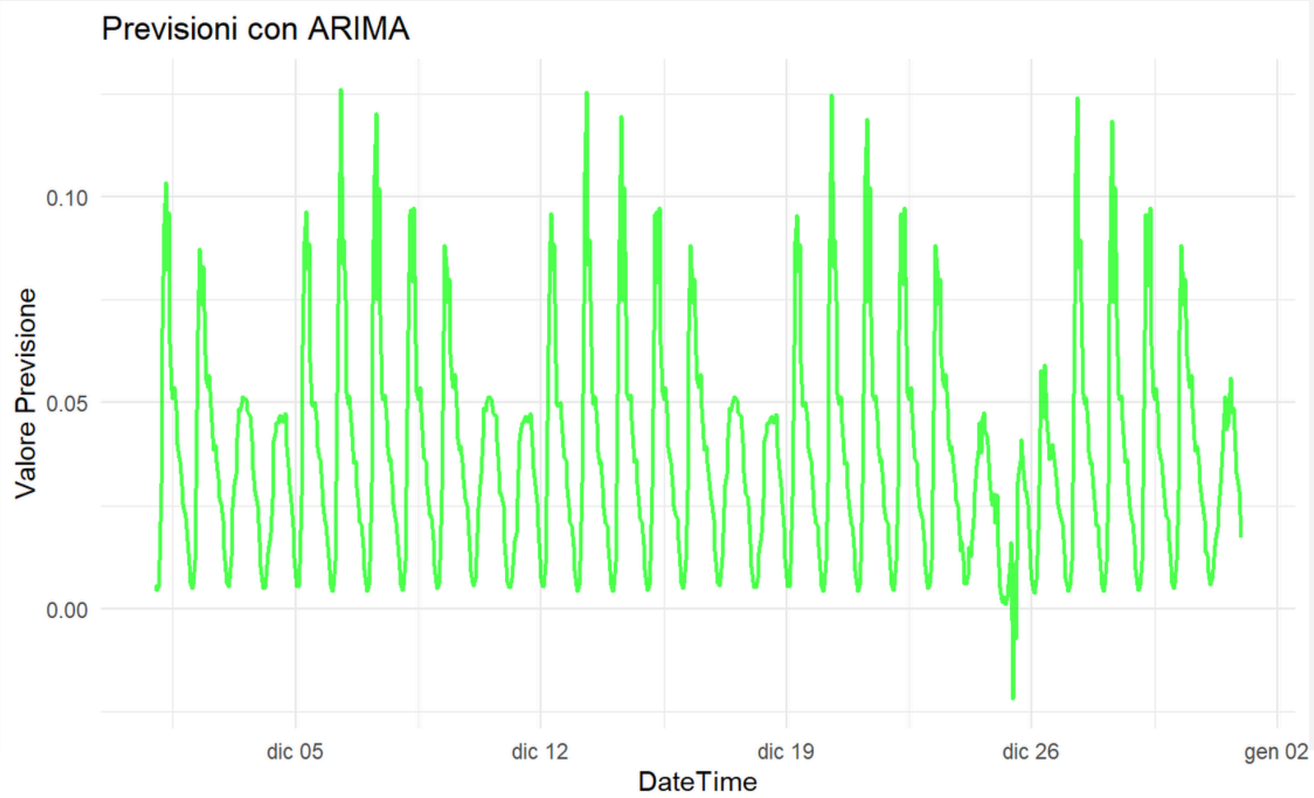
ARIMA MODEL DEPLOYMENT

ARIMA (3,1,1) (0,1,1) (7) with dummies



ARIMA MODEL DEPLOYMENT

ARIMA (3,1,1) (0,1,1) (7) with dummies



UNOBSERVED COMPONENTS MODELS

An iterative approach was used to identify the best-performing UCM model, starting from a simple baseline and gradually adding components, also incorporating holiday dummy variables. Their resulting performances (the average MAE across all 24 series) were then compared.

Without dummies

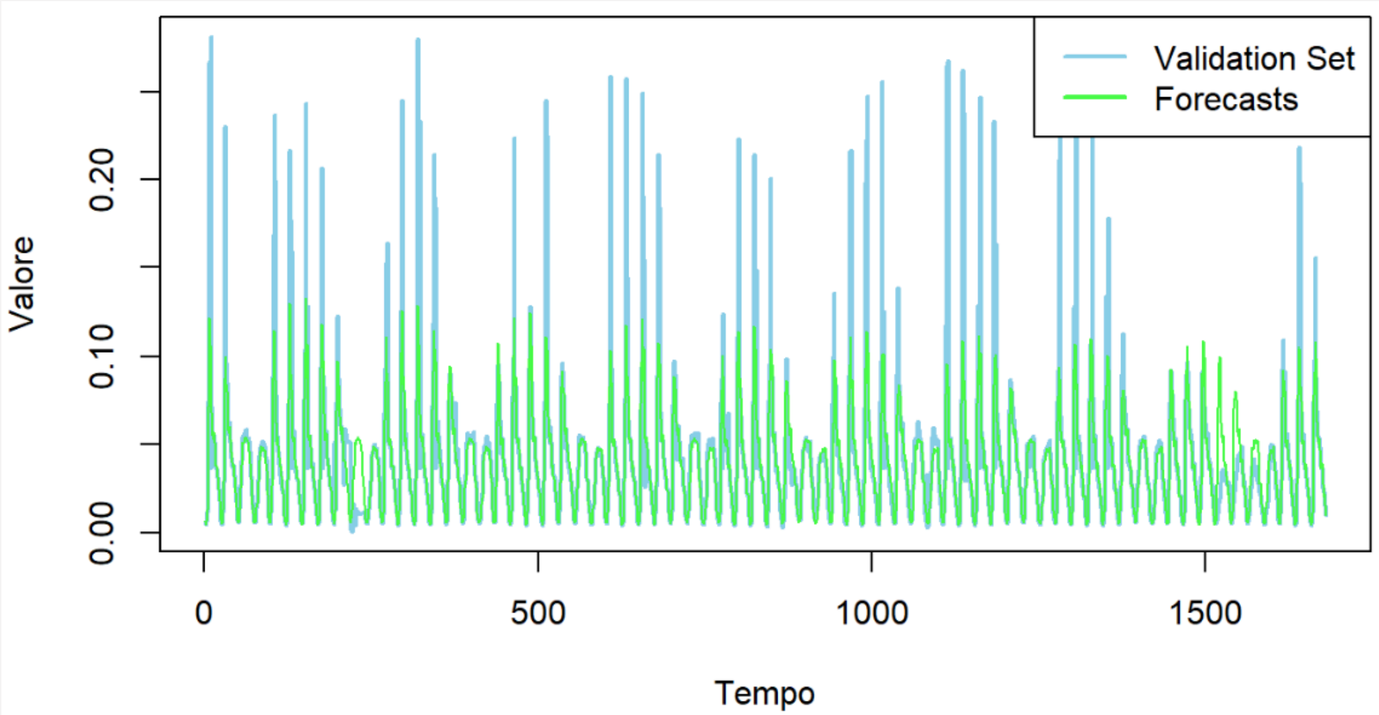
Model	MAE
LLT 7 day dummy seasonality	0.009079
LLT 7 day trigonometric seasonality	0.009193
LLT, 7 day dummy seasonality, 365 day trigonometric seasonality (2 harmonics)	0.008935

With dummies

Model	MAE
LLT 7 day dummy seasonality	0.009044
LLT 7 day trigonometric seasonality	0.009014
LLT, 7 day dummy seasonality, 365 day trigonometric seasonality (2 harmonics)	0.009062

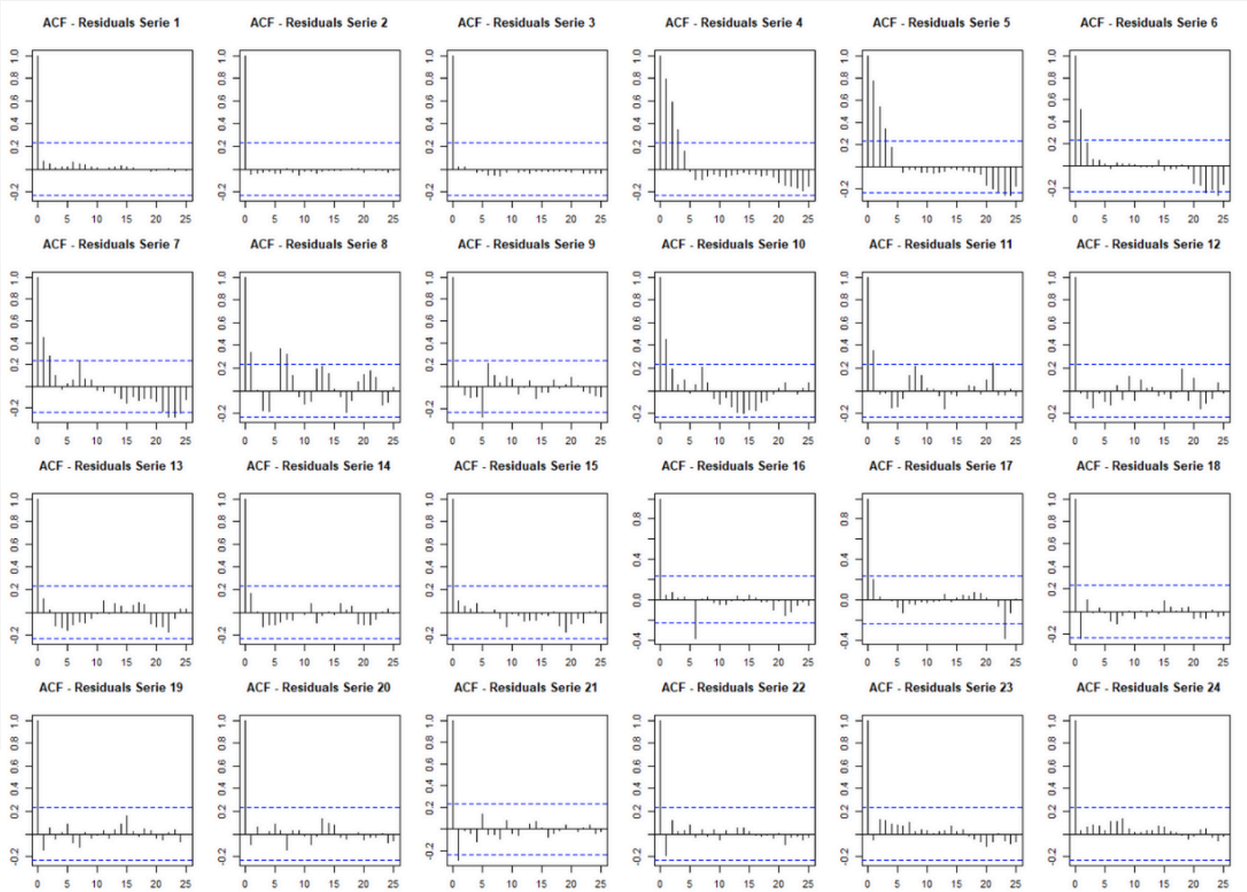
BEST UCM MODEL

LLT, 7 days seasonal dummy seasonality, 365 trigonometric seasonality 2 harmonics



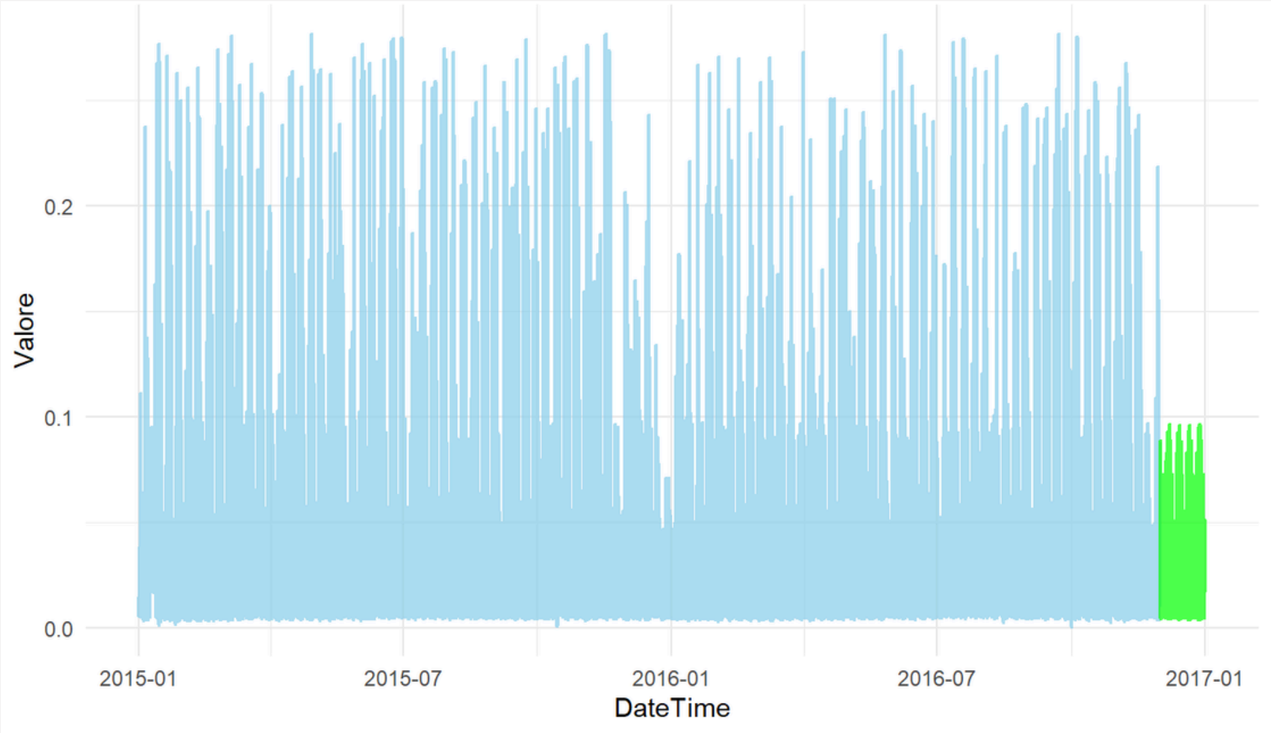
BEST UCM MODEL

LLT, 7 days seasonal dummy seasonality, 365 trigonometric seasonality 2 harmonics



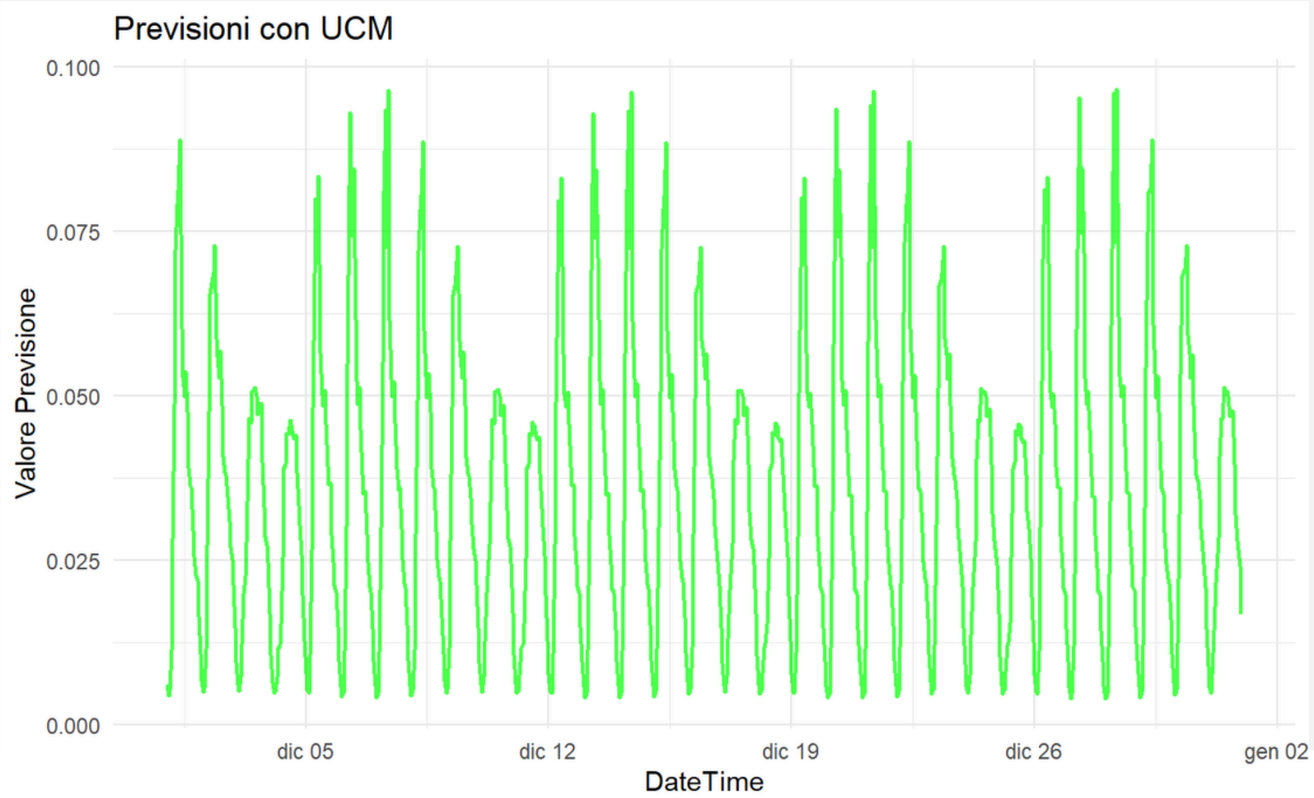
UCM MODEL DEPLOYMENT

LLT, 7 days seasonal dummy seasonality, 365 trigonometric seasonality 2 harmonics



UCM MODEL DEPLOYMENT

LLT, 7 days seasonal dummy seasonality, 365 trigonometric seasonality 2 harmonics



MACHINE LEARNING MODELS

The three most promising ML models were selected and further refined by incorporating holiday dummy variables. Their resulting performances (the average MAE across all 24 series) were then compared.

Without dummies

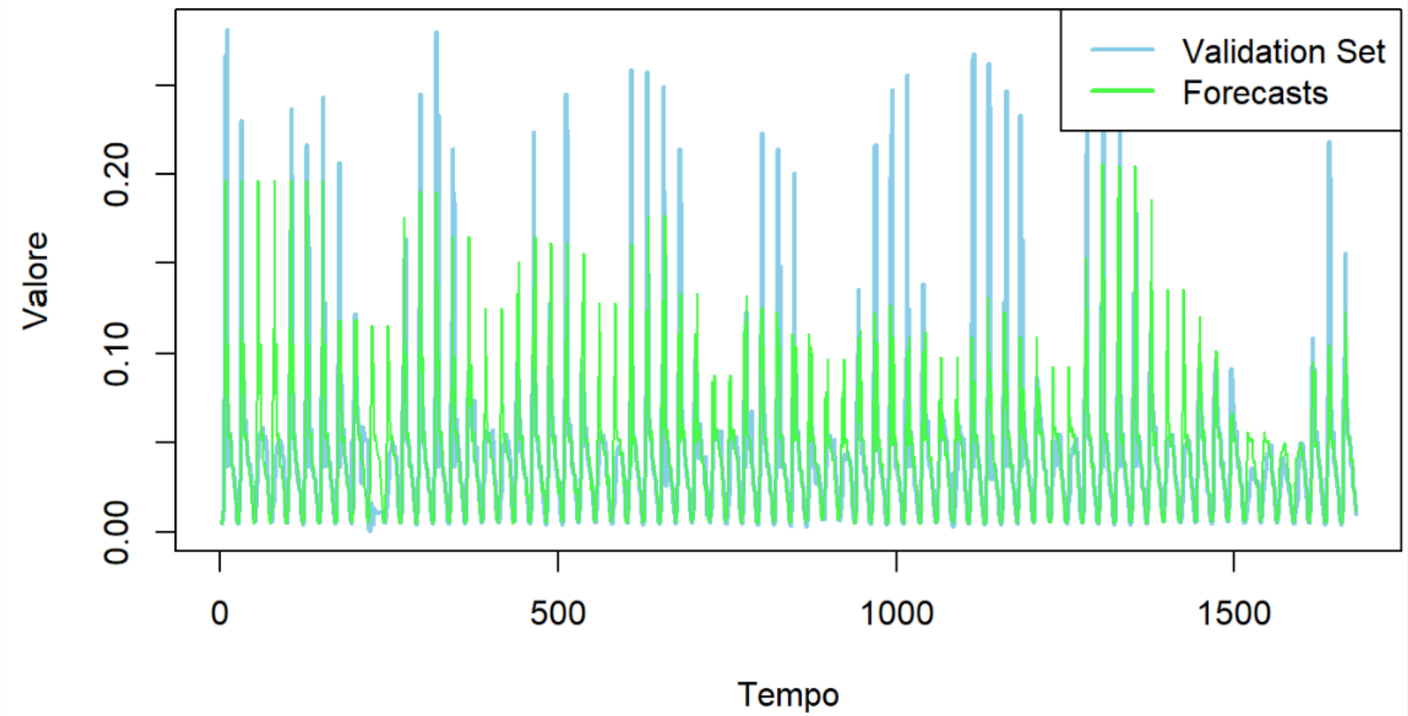
Model	MAE
Random Forest	0.018949
XGBoost	0.022218
K-NN	0.013021

With dummies

Model	MAE
Random Forest	0.018432
XGBoost	0.018701
K-NN	0.013252

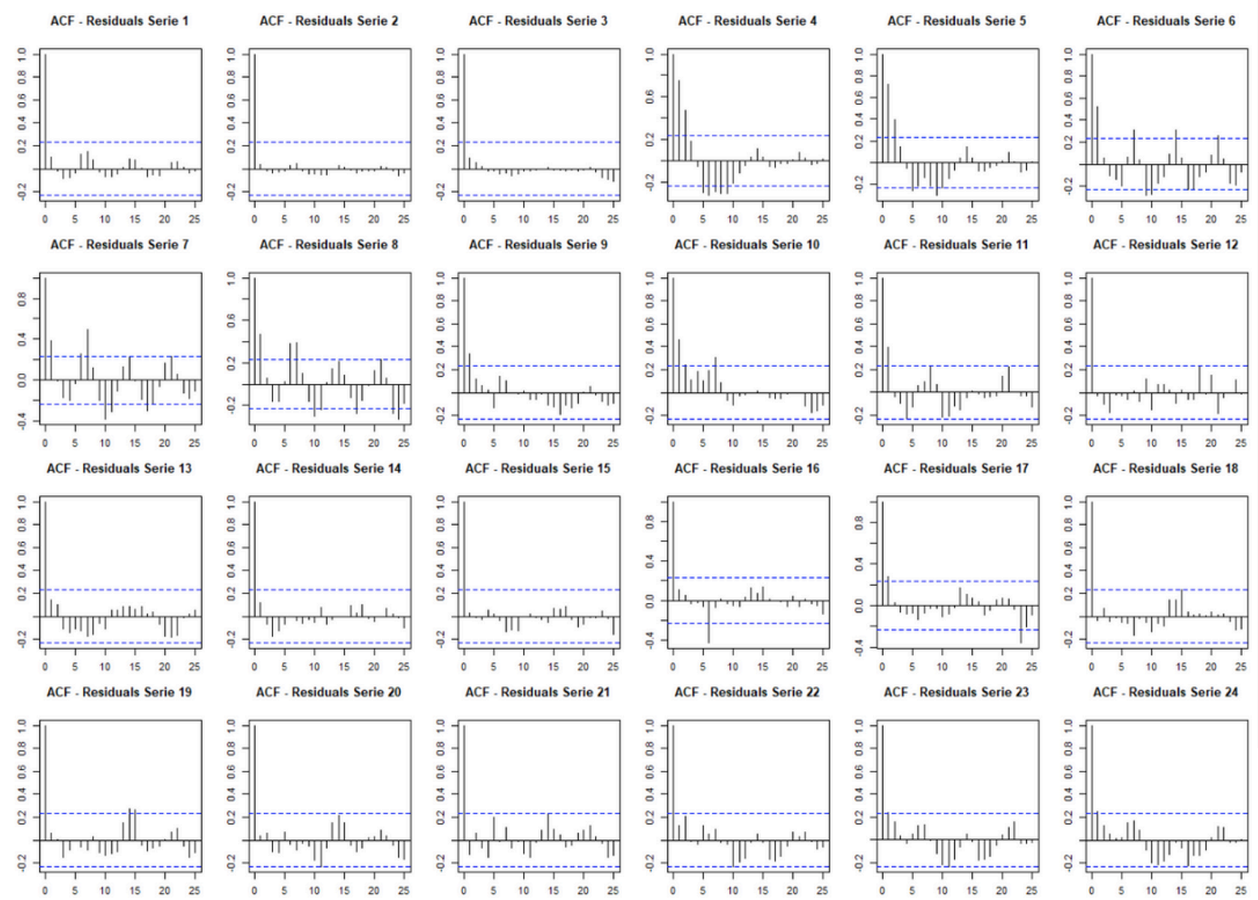
BEST ML MODEL

K-NN



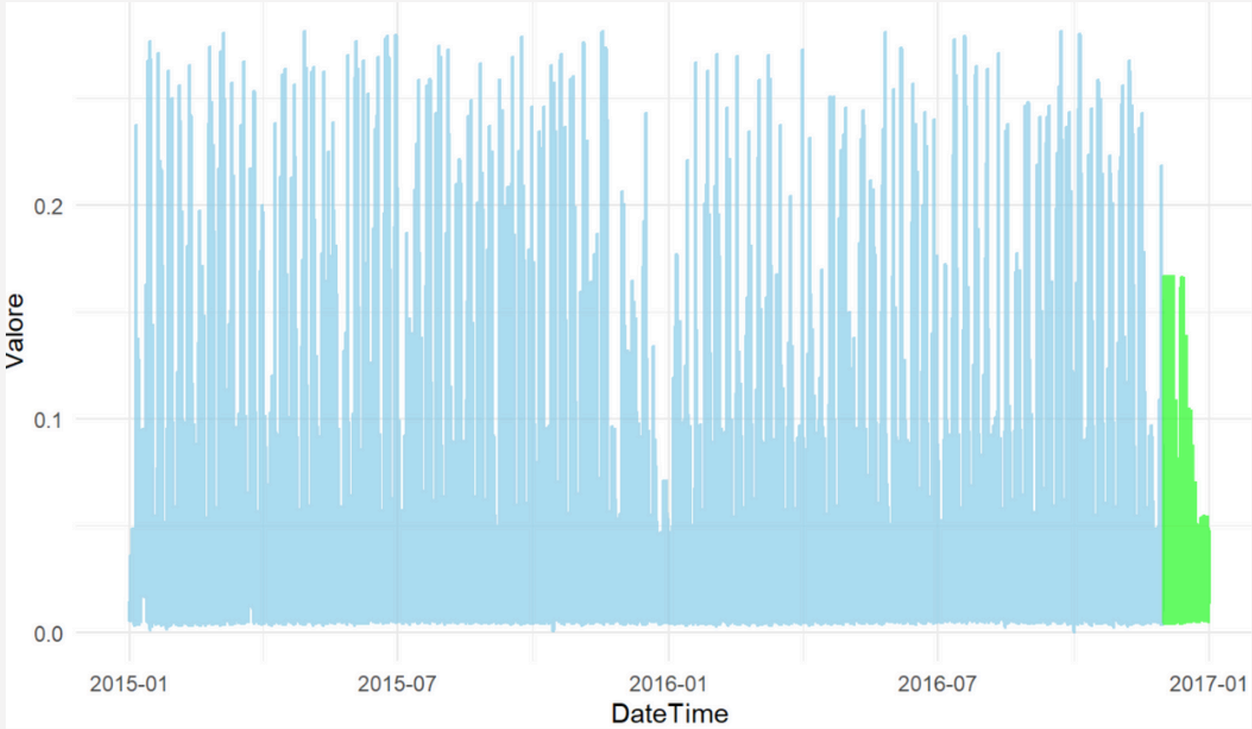
BEST ML MODEL

K-NN



ML MODEL DEPLOYMENT

K-NN



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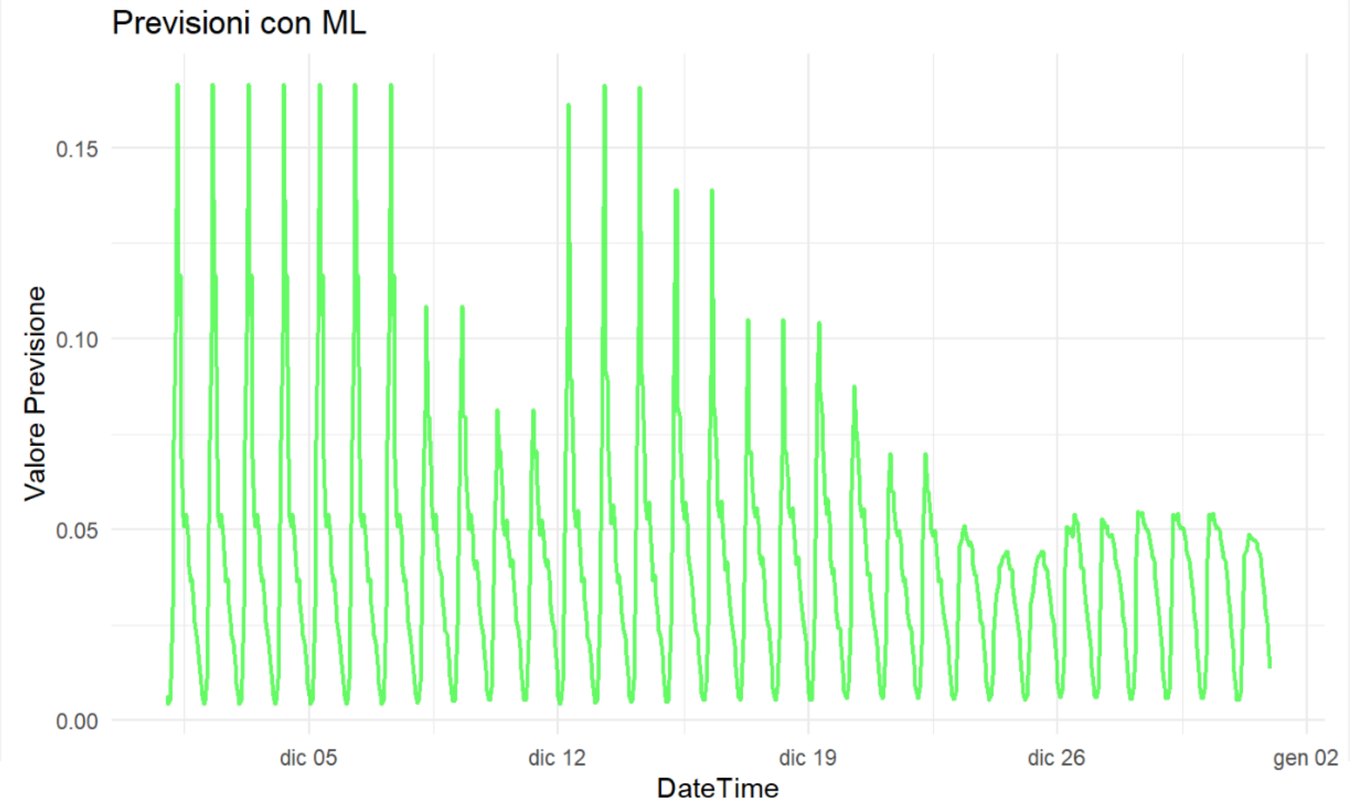
K-NN



FUTURE DEVELOPMENTS

01

PRE-PROCESSING

Explore alternative preprocessing techniques for handling zeros and outliers, such as model-based imputation.

02

COMBINED MODELS

Given the differences across hourly series, combining multiple models tailored to specific hours may improve accuracy of the overall predictions.

03

DUMMY VARIABLES

experimenting with more targeted dummy variables, such as weather conditions or traffic-specific holidays, could enhance model performance.

THANK YOU